

SIMATS ENGINEERING
ASSIGNMENT (COLLECTION 1)

COURSE CODE: CSA0904

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COURSE NAME: Programming in Java

REG NO: 192421410

1. Employee Name List (ArrayList)

Scenario:

A company HR department wants to store the names of employees who attended a training program.

Question :

Write a Java program to: Create an ArrayList<String> Add employee names: Ramesh, Suresh, Priya, Anu, Karthik Display all employee names in the order they were added.

PROGRAM:

```
import java.util.*;
```

```
class Main {
```

```
    public static void main(String[] args) {
```

```
        ArrayList<String> employees = new ArrayList<>();
```

```
        employees.add("Ramesh");
```

```
        employees.add("Suresh");
```

```
        employees.add("Priya");
```

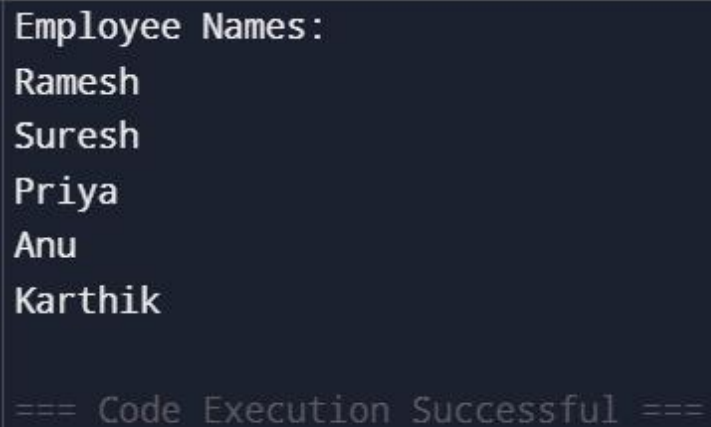
```
        employees.add("Anu");
```

```
        employees.add("Karthik");
```

```
        System.out.println("Employee Names:");
```

```
        for (String name : employees) {  
            System.out.println(name);  
        }  
    }  
}
```

OUTPUT:



```
Employee Names:  
Ramesh  
Suresh  
Priya  
Anu  
Karthik  
  
=== Code Execution Successful ===
```

2. Restaurant Order Management (ArrayList)

Scenario :

A restaurant stores the food items ordered by a customer.

Question :

Write a Java program to: Create an ArrayList<String> Add items: Dosa, Idli, Pongal, Coffee Remove Pongal Display the updated order list and the total number of items.

PROGRAM:

```
import java.util.*;  
  
class Main {  
    public static void main(String[] args) {
```

```
ArrayList<String> orders = new ArrayList<>();

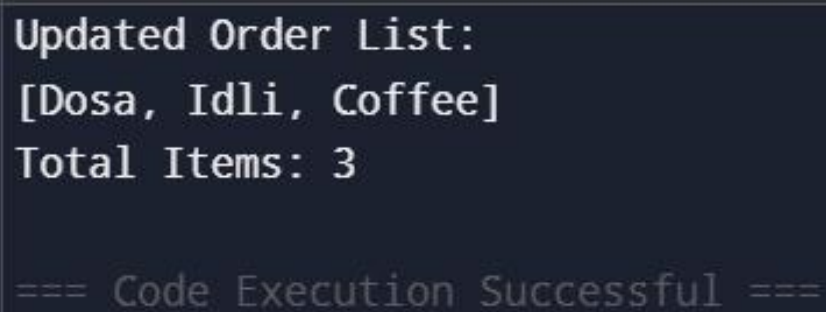
orders.add("Dosa");
orders.add("Idli");
orders.add("Pongal");
orders.add("Coffee");

orders.remove("Pongal");

System.out.println("Updated Order List:");
System.out.println(orders);

System.out.println("Total Items: " + orders.size());
}
}
```

OUTPUT:



```
Updated Order List:
[Dosa, Idli, Coffee]
Total Items: 3

=== Code Execution Successful ===
```

3. Online Course Enrollment (ArrayList)

Scenario:

An online learning platform stores the names of students enrolled in a Java course.

Question :

Write a Java program to: Store 5 student names in an ArrayList.

Check whether Meena is enrolled Print Enrolled if the name exists; otherwise print Not Enrolled.

PROGRAM:

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        ArrayList<String> students = new ArrayList<>();

        students.add("Ravi");
        students.add("Priya");
        students.add("Meena");
        students.add("Anu");
        students.add("Karthik");

        if (students.contains("Meena"))
            System.out.println("Enrolled");
        else
            System.out.println("Not Enrolled");
    }
}
```

OUTPUT:

Enrolled

=== Code Execution Successful ===

4. Unique Email Registration (Set)

Scenario :

A website should not allow duplicate email registrations.

Question:

Write a Java program to: Create a `HashSet<String>`. Add emails: `a@gmail.com`, `b@gmail.com`, `a@gmail.com`, `c@gmail.com`. Display all unique email addresses. Display the total number of unique registered users.

PROGRAM:

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        HashSet<String> emails = new HashSet<>();

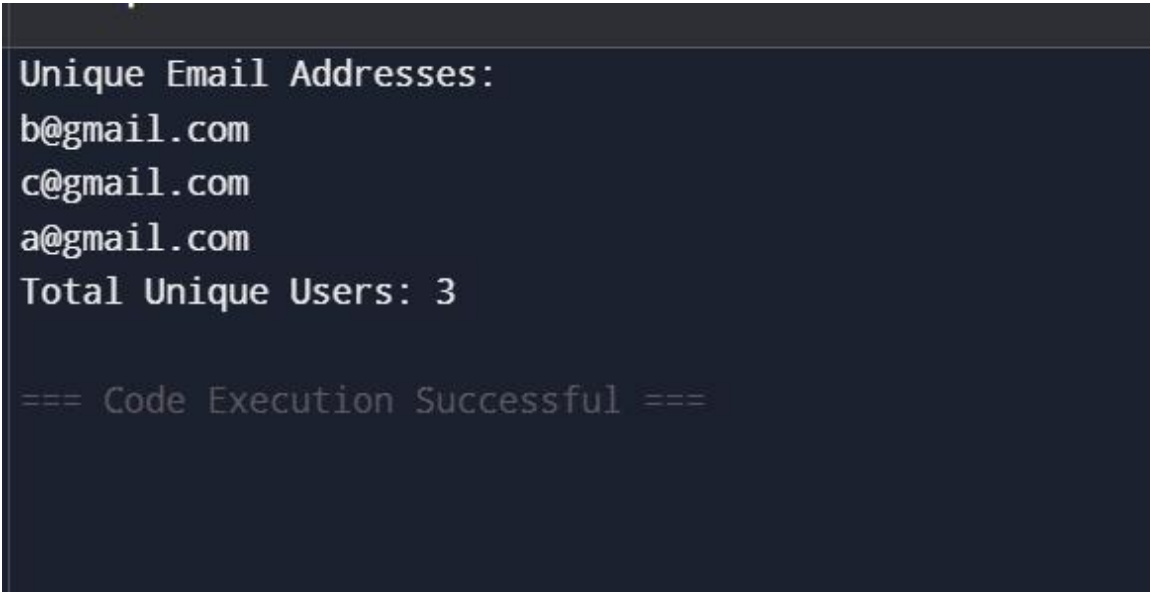
        emails.add("a@gmail.com");
```

```
        emails.add("b@gmail.com");
        emails.add("a@gmail.com");
        emails.add("c@gmail.com");

        System.out.println("Unique Email Addresses:");
        for (String email : emails) {
            System.out.println(email);
        }

        System.out.println("Total Unique Users: " + emails.size());
    }
}
```

OUTPUT:



```
Unique Email Addresses:
b@gmail.com
c@gmail.com
a@gmail.com
Total Unique Users: 3

=== Code Execution Successful ===
```

5. Library Book ID Tracking (Set)

Scenario :

A library stores book ID numbers. The same book ID should not be entered twice.

Question :

Write a Java program to: Create a HashSet<Integer>. Add book IDs: 101, 102, 103, 101, 104. Display all unique book IDs. Print Duplicate Book ID Found when a duplicate ID is added.

PROGRAM:

```
import java.util.*;

class Main {
    public static void main(String[] args) {

        HashSet<Integer> bookIds = new HashSet<>();

        int[] ids = {101, 102, 103, 101, 104};

        for (int id : ids) {

            if (!bookIds.add(id)) {
                System.out.println("Duplicate Book ID Found: " + id);
            }
        }

        System.out.println("Unique Book IDs:");
        for (int id : bookIds) {
            System.out.println(id);
        }
    }
}
```

OUTPUT:

```
Duplicate Book ID Found: 101
```

```
Unique Book IDs:
```

```
101
```

```
102
```

```
103
```

```
104
```

```
=== Code Execution Successful ===
```